

Лабораторная работа № 15. Динамическая маршрутизация

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Цель работы

Настроить динамическую маршрутизацию между территориями организации.

Теоретическое введение

Протокол внутренней маршрутизации Open Shortest Path First (OSPF, RFC 2328) используется на маршрутизаторах для распространения данных маршрутизации внутри одной автономной системы.

Таблица 1: Идентификация маршрутизаторов

IP-адреса	Примечание
10.128.254.0/24	Сеть для идентификации маршрутизаторов
10.128.254.1	msk-donskaya-gw-1
10.128.254.2	msk-q42-gw-1
10.128.254.3	msk-hostel-gw-1
10.128.254.4	sch-sochi-gw-1

Таблица 2: Таблица IP линка 42 квартал–Сочи

IP-адреса	Примечание	VLAN
10.128.255.8/30	Линк между 42 кварталом и Сочи	7
10.128.255.9	msk-q42-gw-1	
10.128.255.10	sch-sochi-gw-1	

Задание

1. Настроить динамическую маршрутизацию по протоколу OSPF на маршрутизаторах msk-donskaya-gw-1, msk-q42-gw-1, msk-hostel-gw-1, sch-sochi-gw-1.
2. Настроить связь сети квартала 42 в Москве с сетью филиала в г. Сочи напрямую.
3. В режиме симуляции отследить движение пакета ICMP с ноутбука администратора сети на Донской в Москве (Laptop-PT admin) до компьютера пользователя в филиале в г. Сочи pc-sochi-1.

4. На коммутаторе провайдера отключить временно vlan 6 и в режиме симуляции убедиться в изменении маршрута прохождения пакета ICMP с ноутбука администратора сети на Донской в Москве (Laptop-PT admin) до компьютера пользователя в филиале в г. Сочи pc-sochi-1.
5. На коммутаторе провайдера восстановить vlan 6 и в режиме симуляции убедиться в изменении маршрута прохождения пакета ICMP с ноутбука администратора сети на Донской в Москве (Laptop-PT admin) до компьютера пользователя в филиале в г. Сочи pc-sochi-1.
6. При выполнении работы необходимо учитывать соглашение об именовании.

Выполнение лабораторной работы

```
msk-donskaya-omabakumova-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-omabakumova-gw-1(config)#router ?
  bgp      Border Gateway Protocol (BGP)
  eigrp    Enhanced Interior Gateway Routing Protocol (EIGRP)
  ospf     Open Shortest Path First (OSPF)
  rip      Routing Information Protocol (RIP)
msk-donskaya-omabakumova-gw-1(config)#router ospf 1
msk-donskaya-omabakumova-gw-1(config-router)#router-id 10.128.254.1
msk-donskaya-omabakumova-gw-1(config-router)#network 10.0.0.0 0.255.255.255 area 0
msk-donskaya-omabakumova-gw-1(config-router)#exit
msk-donskaya-omabakumova-gw-1(config)#^Z
msk-donskaya-omabakumova-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-omabakumova-gw-1#wr mem
Building configuration...
[OK]
msk-donskaya-omabakumova-gw-1#
```

Рис. 1: Включение OSPF на msk-donskaya-omabakumova-gw-1

```
msk-donskaya-omabakumova-gw-1#sh ip ospf
Routing Process "ospf 1" with ID 10.128.254.1
Supports only single TOS(TOS0) routes
Supports opaque LSA
SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
External flood list length 0
  Area BACKBONE(0)
    Number of interfaces in this area is 8
    Area has no authentication
    SPF algorithm executed 1 times
    Area ranges are
    Number of LSA 1. Checksum Sum 0x00312a
    Number of opaque link LSA 0. Checksum Sum 0x000000
    Number of DCbitless LSA 0
    Number of indication LSA 0
    Number of DoNotAge LSA 0
    Flood list length 0

msk-donskaya-omabakumova-gw-1#sh ip ospf neighbor
msk-donskaya-omabakumova-gw-1#sh ip ospf neighbor
```

Рис. 2: Проверка состояния протокола OSPF на маршрутизаторе msk-donskaya-omabakumova-gw-1

Настройка OSPF

```
msk-donskaya-omabakumova-gw-l#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route
```

```
Gateway of last resort is 198.51.100.1 to network 0.0.0.0
```

```
10.0.0.0/8 is variably subnetted, 18 subnets, 4 masks
C    10.128.0.0/24 is directly connected, FastEthernet0/0.3
L    10.128.0.1/32 is directly connected, FastEthernet0/0.3
C    10.128.1.0/24 is directly connected, FastEthernet0/0.2
L    10.128.1.1/32 is directly connected, FastEthernet0/0.2
C    10.128.3.0/24 is directly connected, FastEthernet0/0.101
L    10.128.3.1/32 is directly connected, FastEthernet0/0.101
C    10.128.4.0/24 is directly connected, FastEthernet0/0.102
L    10.128.4.1/32 is directly connected, FastEthernet0/0.102
C    10.128.5.0/24 is directly connected, FastEthernet0/0.103
L    10.128.5.1/32 is directly connected, FastEthernet0/0.103
C    10.128.6.0/24 is directly connected, FastEthernet0/0.104
L    10.128.6.1/32 is directly connected, FastEthernet0/0.104
C    10.128.255.0/30 is directly connected, FastEthernet0/1.5
L    10.128.255.1/32 is directly connected, FastEthernet0/1.5
C    10.128.255.4/30 is directly connected, FastEthernet0/1.6
L    10.128.255.5/32 is directly connected, FastEthernet0/1.6
S    10.129.0.0/16 [1/0] via 10.128.255.2
S    10.130.0.0/16 [1/0] via 10.128.255.6
198.51.100.0/24 is variably subnetted, 2 subnets, 2 masks
C    198.51.100.0/28 is directly connected, FastEthernet0/1.4
L    198.51.100.2/32 is directly connected, FastEthernet0/1.4
S*   0.0.0.0/0 [1/0] via 198.51.100.1
```



```
msk-q42-omabakumova-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-q42-omabakumova-gw-1(config)#router ospf 1
msk-q42-omabakumova-gw-1(config-router)#router-id 10.128.254.2
msk-q42-omabakumova-gw-1(config-router)#network 10.0.0.0 0.255.255.255 area 0
msk-q42-omabakumova-gw-1(config-router)#exit
msk-q42-omabakumova-gw-1(config)#^Z
msk-q42-omabakumova-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

00:10:08: %OSPF-5-ADJCHG: Process 1, Nbr 10.128.254.1 on FastEthernet0/1.5 from LOADING
to FULL, Loading Done

msk-q42-omabakumova-gw-1#wr mem
Building configuration...
[OK]
```

Рис. 4: Включение OSPF на msk-q42-omabakumova-gw-1

```
msk-q42-omabakumova-gw-1#sh ip ospf
Routing Process "ospf 1" with ID 10.128.254.2
Supports only single TOS(TOS0) routes
Supports opaque LSA
SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
External flood list length 0
  Area BACKBONE(0)
    Number of interfaces in this area is 3
    Area has no authentication
    SPF algorithm executed 3 times
    Area ranges are
    Number of LSA 3. Checksum Sum 0x0185e9
    Number of opaque link LSA 0. Checksum Sum 0x000000
    Number of DCbitless LSA 0
    Number of indication LSA 0
    Number of DoNotAge LSA 0
    Flood list length 0
```

Рис. 5: Проверка состояния протокола OSPF на маршрутизаторе msk-q42-omabakumova-gw-1

```
msk-q42-omabakumova-gw-1#sh ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.128.254.1	1	FULL/DR	00:00:38	10.128.255.1	FastEthernet0/1.5

```
msk-q42-omabakumova-gw-1#
```

Рис. 6: Проверка состояния протокола OSPF на маршрутизаторе msk-q42-omabakumova-gw-1

Настройка OSPF

```
msk-q42-omabakumova-gw-1#sh ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address      Interface
10.128.254.1      1    FULL/DR         00:00:38    10.128.255.1 FastEthernet0/1.5

msk-q42-omabakumova-gw-1#
msk-q42-omabakumova-gw-1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.128.255.1 to network 0.0.0.0

10.0.0.0/8 is variably subnetted, 14 subnets, 4 masks
O    10.128.0.0/24 [110/2] via 10.128.255.1, 00:01:42, FastEthernet0/1.5
O    10.128.1.0/24 [110/2] via 10.128.255.1, 00:01:42, FastEthernet0/1.5
O    10.128.3.0/24 [110/2] via 10.128.255.1, 00:01:42, FastEthernet0/1.5
O    10.128.4.0/24 [110/2] via 10.128.255.1, 00:01:42, FastEthernet0/1.5
O    10.128.5.0/24 [110/2] via 10.128.255.1, 00:01:42, FastEthernet0/1.5
O    10.128.6.0/24 [110/2] via 10.128.255.1, 00:01:42, FastEthernet0/1.5
C    10.128.255.0/30 is directly connected, FastEthernet0/1.5
L    10.128.255.2/32 is directly connected, FastEthernet0/1.5
O    10.128.255.4/30 [110/2] via 10.128.255.1, 00:01:42, FastEthernet0/1.5
C    10.129.0.0/24 is directly connected, FastEthernet0/0.201
L    10.129.0.1/32 is directly connected, FastEthernet0/0.201
C    10.129.1.0/24 is directly connected, FastEthernet1/0.202
L    10.129.1.1/32 is directly connected, FastEthernet1/0.202
S    10.129.128.0/17 [1/0] via 10.129.1.2
S*   0.0.0.0/0 [1/0] via 10.128.255.1
```

Рис. 7: Проверка состояния протокола OSPF на маршрутизаторе msk-q42-omabakumova-gw-1

```
msk-hostel-omabakumova-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-hostel-omabakumova-gw-1(config)#router ospf 1
msk-hostel-omabakumova-gw-1(config-router)#router-id 10.128.254.3
msk-hostel-omabakumova-gw-1(config-router)#network 10.0.0.0 0.255.255.255 area 0
msk-hostel-omabakumova-gw-1(config-router)#exit
msk-hostel-omabakumova-gw-1(config)#^Z
msk-hostel-omabakumova-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-hostel-omabakumova-gw-1#wr mem
00:14:55: %OSPF-5-ADJCHG: Process 1, Nbr 10.128.254.2 on Vlan202 from LOADING to FULL,
Loading Done

Building configuration...
[OK]
msk-hostel-omabakumova-gw-1#
```

Рис. 8: Включение OSPF на msk-hostel-omabakumova-gw-1

```
msh-hostel-omabakumova-gw-1#sh ip ospf
Routing Process "ospf 1" with ID 10.128.254.3
Supports only single TOS(TOS0) routes
Supports opaque LSA
SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
External flood list length 0
  Area BACKBONE(0)
    Number of interfaces in this area is 2
    Area has no authentication
    SPF algorithm executed 2 times
    Area ranges are
    Number of LSA 5. Checksum Sum 0x042083
    Number of opaque link LSA 0. Checksum Sum 0x000000
    Number of DCbitless LSA 0
    Number of indication LSA 0
    Number of DoNotAge LSA 0
    Flood list length 0
```

Рис. 9: Проверка состояния протокола OSPF на msk-hostel-omabakumova-gw-1

```
msk-hostel-omabakumova-gw-1#sh ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.128.254.2	1	FULL/DR	00:00:34	10.129.1.1	Vlan202

```
msk-hostel-omabakumova-gw-1#
```

Рис. 10: Проверка состояния протокола OSPF на msk-hostel-omabakumova-gw-1

```
msk-hostel-omabakumova-gw-1#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.129.1.1 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 11 subnets, 2 masks
O       10.128.0.0/24 [110/3] via 10.129.1.1, 00:00:58, Vlan202
O       10.128.1.0/24 [110/3] via 10.129.1.1, 00:00:58, Vlan202
O       10.128.3.0/24 [110/3] via 10.129.1.1, 00:00:58, Vlan202
O       10.128.4.0/24 [110/3] via 10.129.1.1, 00:00:58, Vlan202
O       10.128.5.0/24 [110/3] via 10.129.1.1, 00:00:58, Vlan202
O       10.128.6.0/24 [110/3] via 10.129.1.1, 00:00:58, Vlan202
O       10.128.255.0/30 [110/2] via 10.129.1.1, 00:00:58, Vlan202
O       10.128.255.4/30 [110/3] via 10.129.1.1, 00:00:58, Vlan202
O       10.129.0.0/24 [110/2] via 10.129.1.1, 00:00:58, Vlan202
C       10.129.1.0/24 is directly connected, Vlan202
C       10.129.128.0/24 is directly connected, Vlan301
S*    0.0.0.0/0 [1/0] via 10.129.1.1
```

Рис. 11: Проверка состояния протокола OSPF на msk-hostel-omabakumova-gw-1


```
sch-sochi-omabakumova-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
sch-sochi-omabakumova-gw-1(config)#router ospf 1
sch-sochi-omabakumova-gw-1(config-router)#router-id 10.128.254.4
sch-sochi-omabakumova-gw-1(config-router)#network 10.0.0.0 0.255.255.255 area 0
sch-sochi-omabakumova-gw-1(config-router)#exit
sch-sochi-omabakumova-gw-1(config)#^Z
sch-sochi-omabakumova-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

sch-sochi-omabakumova-gw-1#wr mem
Building configuration...
[OK]
```

Рис. 12: Включение OSPF на sch-sochi-omabakumova-gw-1

```
sch-sochi-omabakumova-gw-1#
00:18:54: %OSPF-5-ADJCHG: Process 1, Nbr 10.128.254.1 on FastEthernet0/0.6 from LOADING to FULL, Loading Done

sch-sochi-omabakumova-gw-1#sh ip ospf
Routing Process "ospf 1" with ID 10.128.254.4
Supports only single TOS(TOS0) routes
Supports opaque LSA
SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
External flood list length 0
  Area BACKBONE(0)
    Number of interfaces in this area is 3
    Area has no authentication
    SPF algorithm executed 3 times
    Area ranges are
    Number of LSA 7. Checksum Sum 0x0640e7
    Number of opaque link LSA 0. Checksum Sum 0x000000
    Number of DCbitless LSA 0
    Number of indication LSA 0
    Number of DoNotAge LSA 0
    Flood list length 0
```

Рис. 13: Проверка состояния протокола OSPF на sch-sochi-omabakumova-gw-1

```
sch-sochi-omabakumova-gw-1#sh ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.120.254.1	1	FULL/DR	00:00:31	10.120.255.5	FastEthernet0/0.6

Рис. 14: Проверка состояния протокола OSPF на sch-sochi-omabakumova-gw-1

```
sch-sochi-omabakumova-gw-1#sh ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address      Interface
10.128.254.1      1    FULL/DR         00:00:31    10.128.255.5 FastEthernet0/0.6

sch-sochi-omabakumova-gw-1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.128.255.5 to network 0.0.0.0

10.0.0.0/8 is variably subnetted, 16 subnets, 3 masks
O    10.128.0.0/24 [110/2] via 10.128.255.5, 00:00:22, FastEthernet0/0.6
O    10.128.1.0/24 [110/2] via 10.128.255.5, 00:00:22, FastEthernet0/0.6
O    10.128.3.0/24 [110/2] via 10.128.255.5, 00:00:22, FastEthernet0/0.6
O    10.128.4.0/24 [110/2] via 10.128.255.5, 00:00:22, FastEthernet0/0.6
O    10.128.5.0/24 [110/2] via 10.128.255.5, 00:00:22, FastEthernet0/0.6
O    10.128.6.0/24 [110/2] via 10.128.255.5, 00:00:22, FastEthernet0/0.6
O    10.128.255.0/30 [110/2] via 10.128.255.5, 00:00:22, FastEthernet0/0.6
C    10.128.255.4/30 is directly connected, FastEthernet0/0.6
L    10.128.255.6/32 is directly connected, FastEthernet0/0.6
O    10.129.0.0/24 [110/3] via 10.128.255.5, 00:00:22, FastEthernet0/0.6
O    10.129.1.0/24 [110/3] via 10.128.255.5, 00:00:22, FastEthernet0/0.6
O    10.129.128.0/24 [110/4] via 10.128.255.5, 00:00:22, FastEthernet0/0.6
C    10.130.0.0/24 is directly connected, FastEthernet0/0.401
L    10.130.0.1/32 is directly connected, FastEthernet0/0.401
C    10.130.1.0/24 is directly connected, FastEthernet0/0.402
L    10.130.1.1/32 is directly connected, FastEthernet0/0.402
S*   0.0.0.0/0 [1/0] via 10.128.255.5
```

Рис. 15: Проверка состояния протокола OSPF на sch-sochi-omabakumova-gw-1

Настройка линка 42-й квартал–Сочи

```
provider-omabakumova-sw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
provider-omabakumova-sw-1(config)#vlan 7
provider-omabakumova-sw-1(config-vlan)#name q42-sochi
provider-omabakumova-sw-1(config-vlan)#exit
provider-omabakumova-sw-1(config)#interface vlan7
provider-omabakumova-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan7, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan7, changed state to up

provider-omabakumova-sw-1(config-if)#no shutdown
provider-omabakumova-sw-1(config-if)#exit
provider-omabakumova-sw-1(config)#^Z
provider-omabakumova-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

provider-omabakumova-sw-1#wr mem
Building configuration...
[OK]
```

Рис. 16: Настройка интерфейсов provider-omabakumova-sw-1

```
!
interface Vlan4
 no ip address
!
interface Vlan5
 no ip address
!
interface Vlan6
 no ip address
!
interface Vlan7
 no ip address
!
!
!
!
line con 0
```

Рис. 17: Результаты настройки интерфейсов provider-omabakumova-sw-1

Настройка линка 42-й квартал–Сочи

```
provider-omabakumova-sw-1#sh vlan
```

VLAN	Name	Status	Ports
1	default	active	Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
4	nat	active	
5	q42	active	
6	sochi	active	
7	q42-sochi	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Transl	Trans2
1	enet	100001	1500	-	-	-	-	-	0	0

Рис. 18: Результаты настройки интерфейсов provider-omabakumova-sw-1

Настройка линка 42-й квартал–Сочи

```
msk-q42-omabakumova-gw-1(config)#interface f0/1.7
msk-q42-omabakumova-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/1.7, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.7, changed state to up

msk-q42-omabakumova-gw-1(config-subif)#encapsulation dot1Q 7
msk-q42-omabakumova-gw-1(config-subif)#ip address 10.120.255.9 255.255.255.252
msk-q42-omabakumova-gw-1(config-subif)#description sochi
msk-q42-omabakumova-gw-1(config-subif)#exit
msk-q42-omabakumova-gw-1(config)#^Z
msk-q42-omabakumova-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-q42-omabakumova-gw-1#wr mem
Building configuration...
[OK]
```

Рис. 19: Настройка msk-q42-omabakumova-gw-1

Настройка линка 42-й квартал-Сочи

```
sch-sochi-omabakumova-sw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
sch-sochi-omabakumova-sw-1(config)#vlan 7
sch-sochi-omabakumova-sw-1(config-vlan)#name q42-sochi
sch-sochi-omabakumova-sw-1(config-vlan)#exit
sch-sochi-omabakumova-sw-1(config)#interface vlan7
sch-sochi-omabakumova-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan7, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan7, changed state to up

sch-sochi-omabakumova-sw-1(config-if)#no shutdown
sch-sochi-omabakumova-sw-1(config-if)#exit
sch-sochi-omabakumova-sw-1(config)#^Z
sch-sochi-omabakumova-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

sch-sochi-omabakumova-sw-1#wr mem
Building configuration...
^C
```

Рис. 20: Настройка sch-sochi-omabakumova-sw-1

Настройка линка 42-й квартал–Сочи

```
sch-sochi-omabakumova-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
sch-sochi-omabakumova-gw-1(config)#interface f0/0.7
sch-sochi-omabakumova-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.7, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.7, changed state to up

sch-sochi-omabakumova-gw-1(config-subif)#encapsulation dot1Q 7
sch-sochi-omabakumova-gw-1(config-subif)#ip address 10.128.255.10 255.255.255.252
sch-sochi-omabakumova-gw-1(config-subif)#description q42
sch-sochi-omabakumova-gw-1(config-subif)#exit
00:32:13: %OSPF-5-ADJCHG: Process 1, Nbr 10.128.254.2 on FastEthernet0/0.7 from LOADING
to FULL, Loading Done

sch-sochi-omabakumova-gw-1(config)#^Z
sch-sochi-omabakumova-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

sch-sochi-omabakumova-gw-1#wr mem
Building configuration...
[OK]
```

Рис. 21: Настройка sch-sochi-omabakumova-gw-1

```
sch-sochi-omabakumova-gw-l#sh ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.128.254.1	1	FULL/DR	00:00:31	10.128.255.5	FastEthernet0/0.6
10.128.254.2	1	FULL/DR	00:00:34	10.128.255.9	FastEthernet0/0.7

```
sch-sochi-omabakumova-gw-l#
```

Рис. 22: Статус всех соседей в заданном сегменте

Отправление пакетов

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection (Default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::120:F7FF:FE08:555B
    IPv4 Address. . . . .: 10
    IPv4 Address. . . . .: 10.120.6.200
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: 10
    Default Gateway . . . . .: 10.120.6.1

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv4 Address. . . . .: 0
    IPv4 Address. . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: 0
    Default Gateway . . . . .: 0.0.0.0

C:\>ping 10.130.0.200

Pinging 10.130.0.200 with 32 bytes of data:

Request timed out.
Reply from 10.130.0.200: bytes=32 time=1ms TTL=126
Reply from 10.130.0.200: bytes=32 time=1ms TTL=126
Reply from 10.130.0.200: bytes=32 time=1ms TTL=126

Ping statistics for 10.130.0.200:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milliseconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 10.130.0.200

Pinging 10.130.0.200 with 32 bytes of data:

Reply from 10.130.0.200: bytes=32 time=1ms TTL=126
Reply from 10.130.0.200: bytes=32 time=1ms TTL=126
Reply from 10.130.0.200: bytes=32 time=1ms TTL=126
Reply from 10.130.0.200: bytes=32 time=1ms TTL=126

Ping statistics for 10.130.0.200:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milliseconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>tracert 10.130.0.200

Tracing route to 10.130.0.200 over a maximum of 30 hops:

  0  0 ms  0 ms  0 ms  10.120.6.1
  1  0 ms  0 ms  0 ms  10.120.355.6
  2  0 ms  0 ms  0 ms  10.130.0.200

Trace complete.
```

Рис. 23: Конфигурация, пинг и вывод маршрута

Отправление пакетов

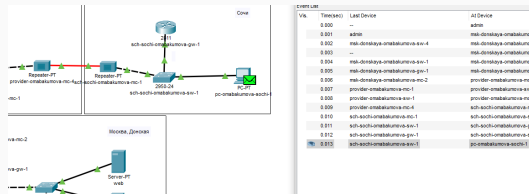


Рис. 24: Пакет успешно дошел по маршруту

```
provider-omabakumova-sw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
provider-omabakumova-sw-1(config)#no vlan 6
provider-omabakumova-sw-1(config)#
%LINK-3-UPDOWN: Interface Vlan6, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan6, changed state to down
provider-omabakumova-sw-1(config)#
```

Рис. 25: Отключение vlan6

Отправление пакетов

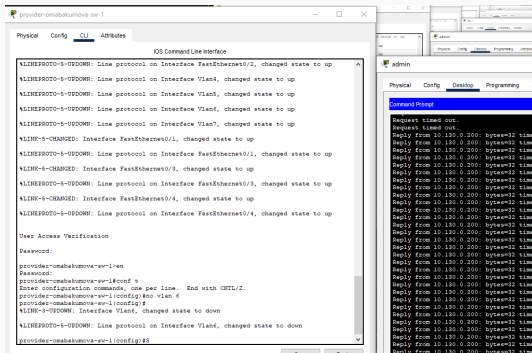


Рис. 26: Изменение маршрута

Отправление пакетов

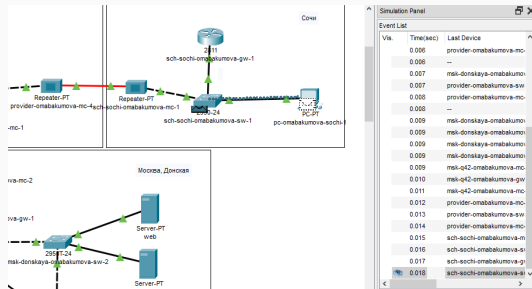


Рис. 27: Пакет успешно дошел по измененному маршруту


```
C:\>tracert 10.130.0.200

Tracing route to 10.130.0.200 over a maximum of 30 hops:

  1  6 ms      6 ms      6 ms      10.128.6.1
  2  18 ms     18 ms     18 ms     10.128.255.2
  3  32 ms     32 ms     32 ms     10.128.255.10
  4  36 ms     36 ms     36 ms     10.130.0.200

Trace complete.
```

Рис. 28: Таблица маршрутизации с учетом отключения vlan6

Отправление пакетов

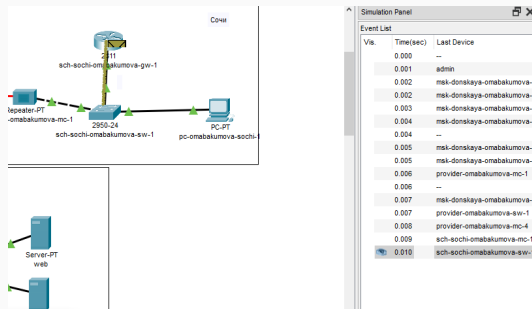


Рис. 29: Маршрут восстановлен, пакет успешно дошел

Выводы

В процессе выполнения лабораторной работы я настроила динамическую маршрутизацию между территориями организации.